Chapter 2: Graph and network optimization

Many decision-making problems can be formulated in terms of graphs and networks.

Examples

- network design (communication, electrical, water,. . .),
- transportation and distribution problems,
- location problems (services and facilities),
- project planning, resource management,
- timetable scheduling,
- production planning, . . .

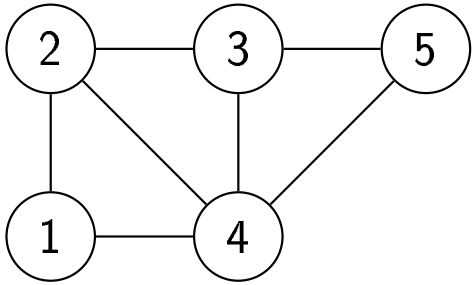
2.1 Graphs

- A graph is a pair $G = (N, E)$, with N a set of nodes (vertices) and $E \subseteq N \times N$ a set of pairs of nodes.
- Undirected $G = (N, E)$ with E set of edges $\{i, j\}$ for some $i, j \in N$ (unordered pairs of nodes).

  Directed $G = (N, A)$ with A set of arcs (i, j) for some $i, j \in N$ (ordered pairs of nodes).

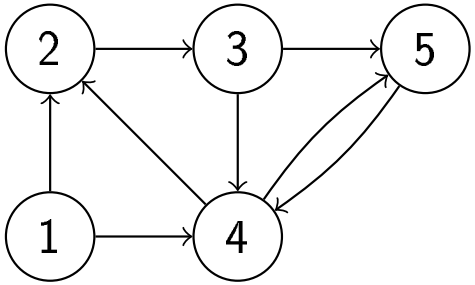
Example: A road network which connects n cities can be modelled by a graph (nodes $\leftrightarrow$ cities, arcs $\leftrightarrow$ connections).

Undirected graph:



$N = \{1, 2, 3, 4, 5\}$
$E = \{\{1, 2\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{3, 5\}, \{4, 5\}\}$

Directed graph:

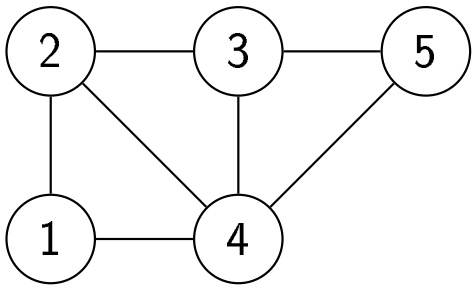


$N = \{1, 2, 3, 4, 5\}$
$E' = \{(1, 2), (1, 4), (2, 3), (4, 2), (3, 4), (3, 5), (4, 5), (5, 4)\}$

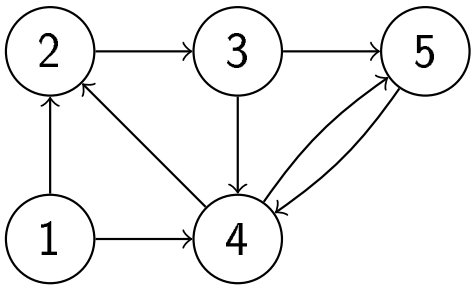
- Two nodes are adjacent if they are connected by an edge.
- An edge e is incident in a node v if v is an endpoint of e.
- Undirected graphs: The degree of a node is the number of incident edges

  Directed graphs: The in-degree (out-degree) of a node is the number of arcs that have it as head (tail).

Examples



Nodes 1 and 2 are adjacent (unlike nodes 1 and 3)
Edge $\{1, 2\}$ is incident in nodes 1 and 2
Node 4 has degree 4



Node 1 has in-degree 0, out-degree 2

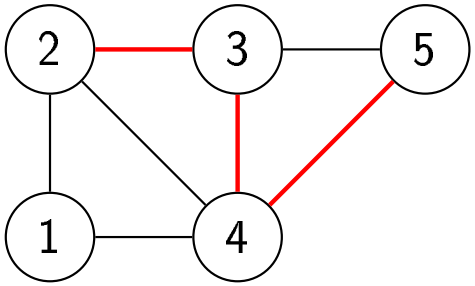
- A path from $v_1 \in N$ to $v_k \in N$ is a sequence of consecutive edges

$$p = \langle \{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{k-1}, v_k\} \rangle$$

  with $\{v_l, v_{l+1}\} \in E$ for $l = 1, \ldots, k-1$.

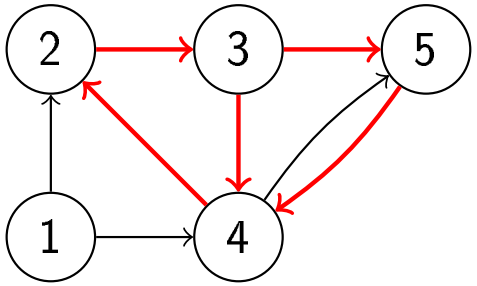
  A directed path from $v_1 \in N$ to $v_k \in N$ is a sequence of consecutive arcs $(v_l, v_{l+1}) \in E$, for $l = 1, \ldots, k-1$.
- Nodes u and v are connected if there is a path connecting them.
- $G = (N, E)$ is connected if u, v are connected, for every $u, v \in N$.
- $G = (N, A)$ is strongly connected if u, v are connected by a directed path, for every $u, v \in N$.

Examples: Connected graph



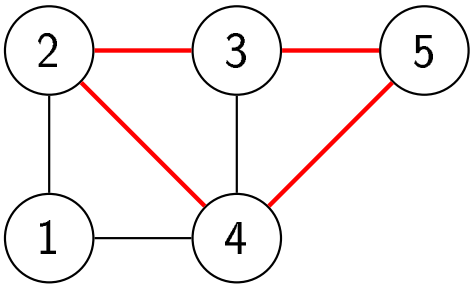
$\langle \{2,3\}, \{3,4\}, \{4,5\} \rangle$ is a path from node 2 to node 5
Nodes 2 and 5 are connected

Not strongly connected graph

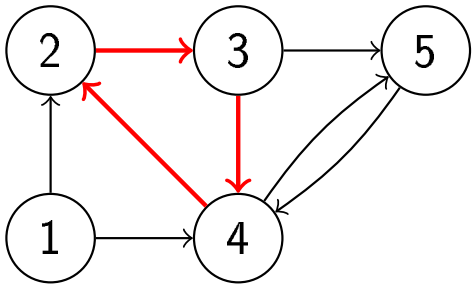


$\langle (3,5), (5,4), (4,2), (2,3), (3,4) \rangle$ is a directed path from node 3 to node 4

- A cycle (circuit) is a (directed) path with $v_1 = v_k$.



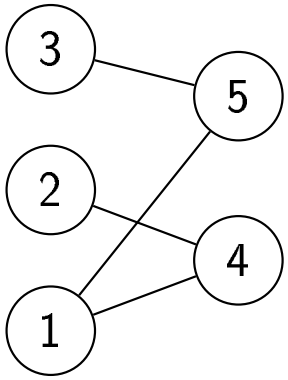
$\langle\{2,3\},\{3,5\},\{5,4\},\{4,2\}\rangle$ is a cycle



$\langle(2,3),(3,4),(4,2)\rangle$ is a circuit

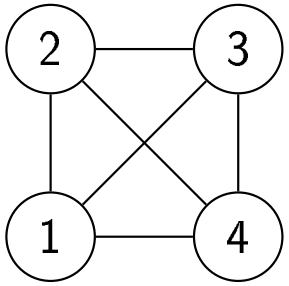
- $G = (N, E)$ is bipartite if there is a partition $N = N_1 \cup N_2$ ($N_1 \cap N_2 = \emptyset$) such that no edge connects nodes in the same subset.
- $G = (N, E)$ is complete if $E = \{\{v_j, v_j\} : v_i, v_j \in N \ \wedge i \leq j\}$.

Bipartite graph:



$N_1 = \{1, 2, 3\}$ and $N_2 = \{4, 5\}$

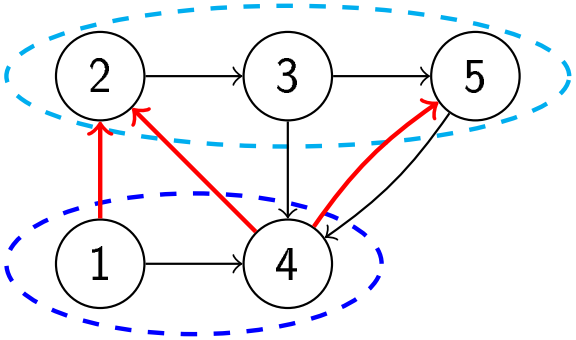
Complete undirected graph:

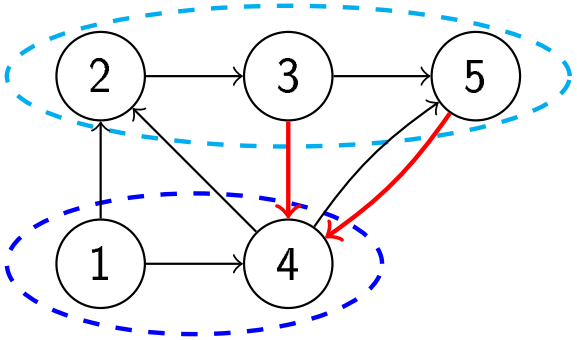


- Given a directed graph $G = (N, A)$ and $S \subset N$, the outgoing cut induced by S is
$$\delta^+(S) = \{(u, v) \in A : u \in S \ \wedge v \in N \backslash S\},$$

  the incoming cut induced by S is
$$\delta^-(S) = \{(u, v) \in A : v \in S \ \wedge u \in N \backslash S\}.$$

Example





$\delta^+(\{1, 4\}) = \{(1, 2), (4, 2), (4, 5)\}$
$S = \{1, 4\}$
$N \backslash S = \{2, 3, 5\}$

$\delta^-(\{1, 4\}) = \{(3, 4), (5, 4)\}$
$S = \{1, 4\}$
$N \backslash S = \{2, 3, 5\}$

For an undirected graph $G = (N, E)$ and $S \subset N$, the cut induced by S is

$$\delta(S) = \{\{u, v\} \in E : u \in S \ \wedge v \in N \backslash S\}.$$

Property:

- An undirected graph $G = (N, E)$ with n nodes has $m \leq \frac{n(n-1)}{2}$ edges.
- A directed graph $G = (N, A)$ with n nodes has $m \leq n(n-1)$ arcs.

G is dense if $m \approx n^2$ and sparse if $m \ll n^2$.

Graph representation

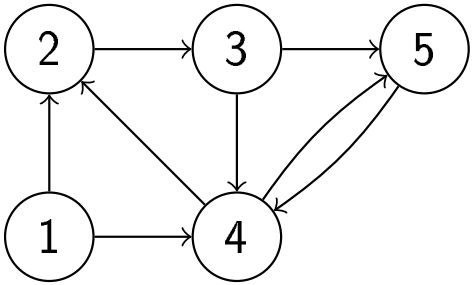
For dense directed graphs, $n \times n$ adjacency matrix:

$$a_{ij} = 1 \text{ if } (i,j) \in A \quad \text{and} \quad a_{ij} = 0 \text{ otherwise.}$$

For sparse directed graphs, list of successors (or predecessors) for each node.

Similar for undirected graphs.

Example



$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \qquad \begin{aligned} S(1) &= \{2,4\} \\ S(2) &= \{3\} \\ S(3) &= \{4,5\} \\ S(4) &= \{2,5\} \\ S(5) &= \{4\} \end{aligned}$$

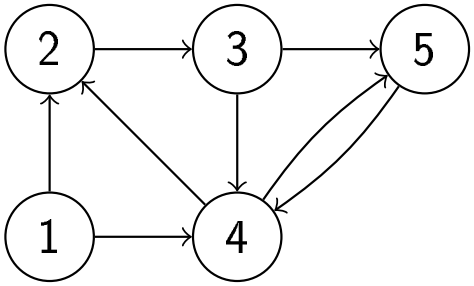
2.1.1 Graph reachability problem

Given a directed graph $G = (N, A)$ and a node s, determine all the nodes that are reachable from s.

Input: $G = (N, A)$, described by the successor lists, and $s \in N$.
Output: Subset $M \subseteq N$ of nodes of G reachable from s.

Efficient algorithm uses a "queue" Q containing the nodes reachable from s and not yet processed (First-In First-Out policy).

Example



$Q = \{2\}$ $\quad M = \emptyset$

$Q = \{\cancel{2}\}$ $\quad M = \{2\}$

$Q = \{\cancel{3}\}$ $\quad M = M \cup \{3\}$

$Q = \{\cancel{4}, 5\}$ $\quad M = M \cup \{4\}$

$Q = \{\cancel{5}\}$ $\quad M = M \cup \{5\}$

$Q = \emptyset$

$M = \{2, 3, 4, 5\}$ is the set of nodes labeled, i.e., reachable from $s = 2$.

Algorithm for the graph reachability problem

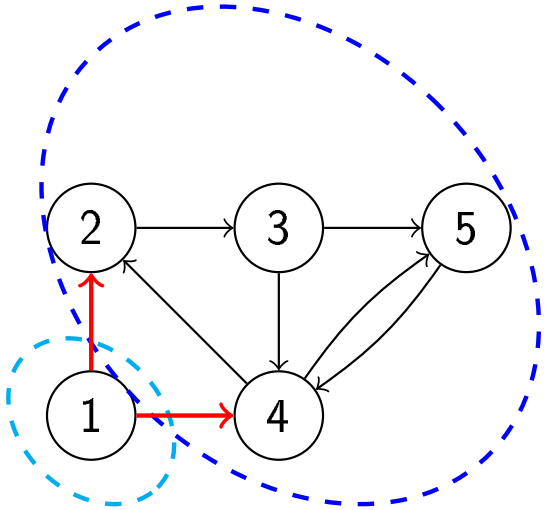
- Input: Graph $G = (N, A)$, described by the successor lists and node $s \in N$.
- Output: Subset $M \subseteq N$ of nodes of G reachable from s.

Algorithm 1: Graph reachability problem

```
Q ← {s}; M ← ∅
while Q ≠ ∅ do
    u ← node in Q; Q ← Q\{u}
    M ← M ∪ {u}          /* label u                    */
    for v ∈ S(u) do
        if v ∉ M and v ∉ Q then  Q ← Q ∪ {v}
```

If Q is managed as a FIFO queue, the nodes are explored by breadth-first search.

Graph reachability problem



- The algorithm stops when $\delta^+(M) = \emptyset$.
- $\delta^-(M)$ is the set of arcs with head node in M and tail node in $N \backslash M$.

2.1.2 Complexity of algorithms

- An algorithm for a problem is a sequence of instructions that allows to solve any of its instances.
- The execution time of an algorithm depends on:
  - the instance,
  - the computer.
- Evaluate the complexity of the algorithm as a function of the size of the instance independently from the hardware.
- Consider the number of elementary operations (e.g., arithmetic operations, comparisons, memory accesses, ...) and assume all have the same cost.

Example

1. The dot product of $\underline{a}, \underline{b} \in \mathbb{R}^n$ requires $2n - 1$ elementary operations.
2. Given two matices $A, B \in \mathbb{R}^{n \times n}$, the product AB requires $(2n - 1)n^2$ elementary operations.

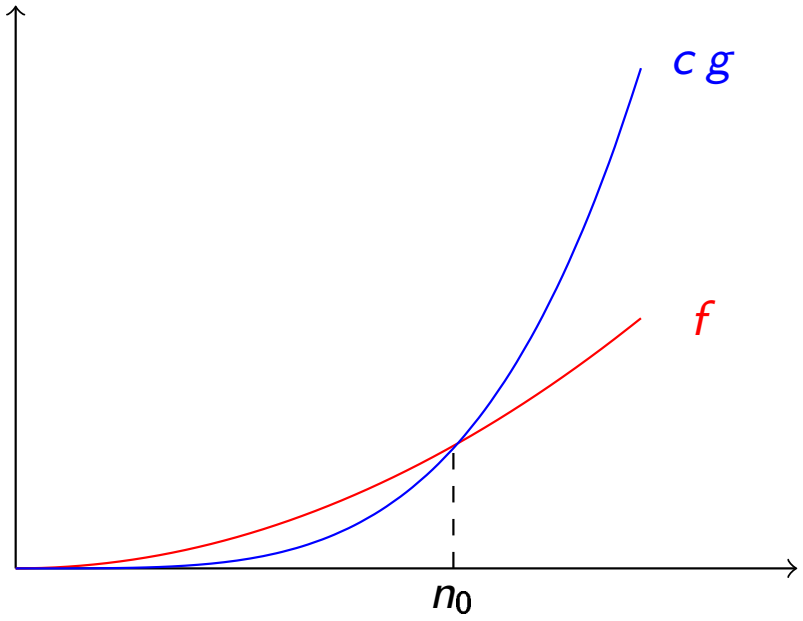
Complexity of algorithms

- Since it is often hard to determine the exact number of elementary operations (as a function of the instance size), we consider the asymptotic number of elementary operations (speed of growth) in the worst case.
- We look for a function $f(n)$ such that, for n sufficiently large,

  the number of elementary operations needed to solve any instance of size at most n $\leq f(n)$.

Big-O notation

A function f is order of g, written $f(n) = O(g(n))$, if $\exists c > 0$ such that $f(n) \leq c\, g(n)$ (asymptotically), for n sufficiently large.



Examples

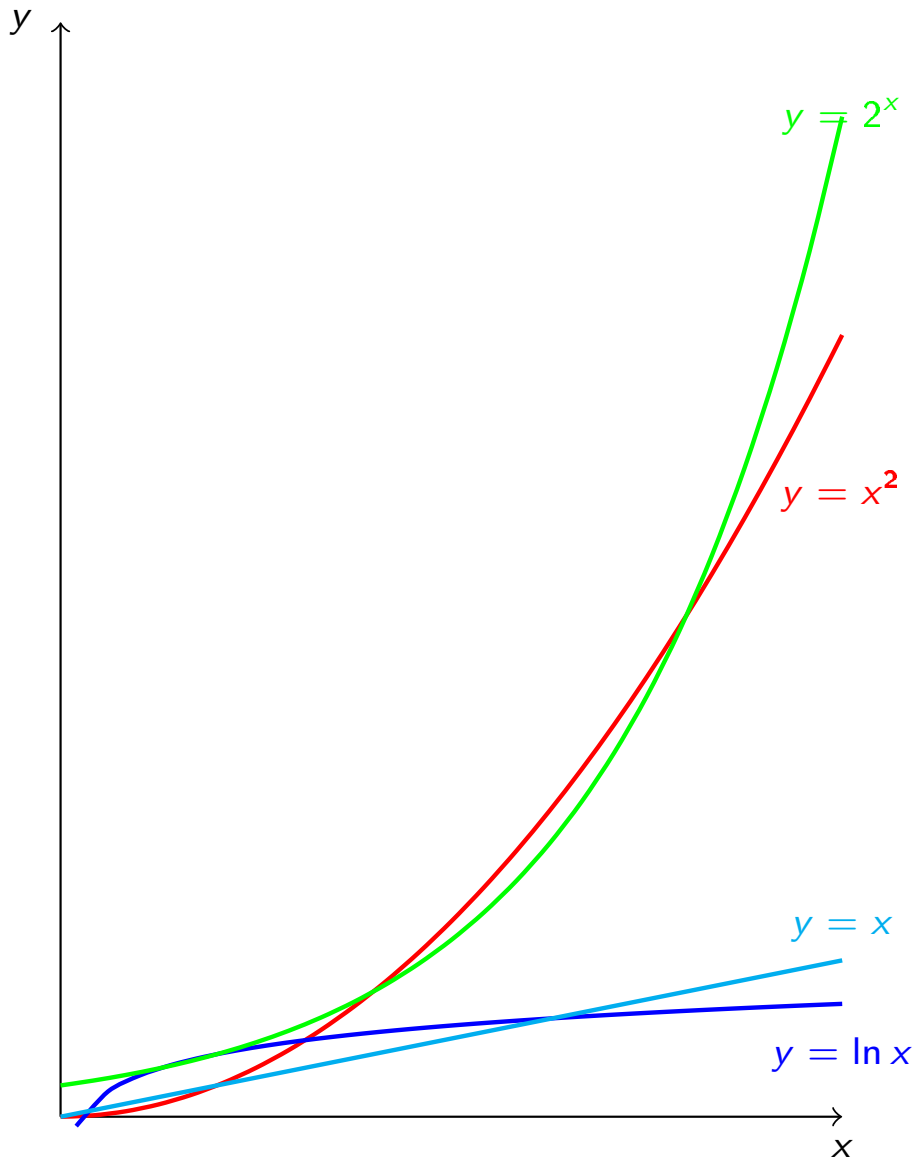
- $4n^3 - n^2 + n$ is of $O(n^3)$
- $\frac{n(n-1)}{2} + 1$ is of $O(n^2)$

Complexity of algorithms

Distinction based on worst-case order of complexity:

- polynomial: $O(n^d)$ for a given constant d
- exponential: $O(2^n)$

An $O(n^d)$ algorithm with large d (e.g. $d = 8$) is not efficient in practice!



Complexity of algorithms

Distinction based on worst-case order of complexity:

- polynomial: $O(n^d)$, $d \in \mathbb{R}$
- exponential: $O(2^n)$

Assuming that 1 microsecond is needed per elementary operation:

n	n^2	2^n
1	0.000001 secs	0.000002 secs
10	0.0001 secs	0.001024 secs
20	0.0004 secs	1.048576 secs
30	0.0009 secs	18 mins
40	0.0016 secs	13 days
50	0.0025 secs	36 years
60	0.0036 secs	366 centuries

Complexity of the reachability algorithm

At each iteration of the `while` loop:

- Select one node $u \in Q$, extract it from Q and insert it in M.
- For all nodes v directly reachable from u and not already in M or Q, insert v in Q.

Since each node u is inserted in Q at most once and each arc (u, v) is considered at most once, the overall complexity is

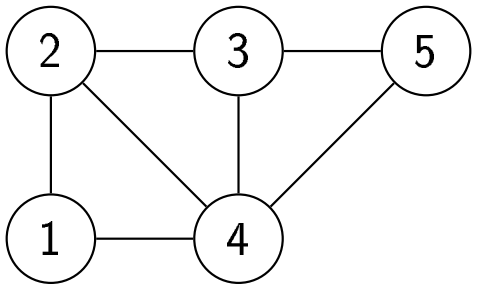
$$O(n + m), \text{ where } n = |N|, m = |A|.$$

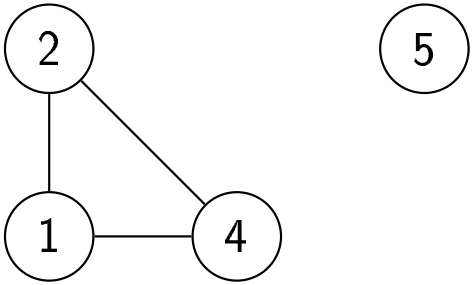
Note: For dense graphs, $m = O(n^2)$.

2.1.3 Subgraphs and trees

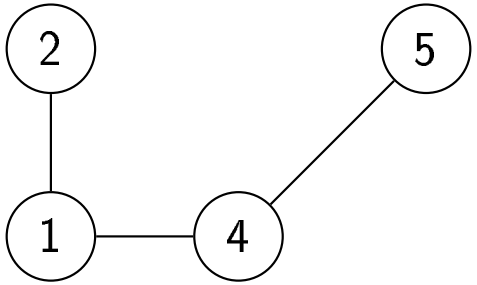
- $G' = (N', E')$ is a subgraph of $G = (N, E)$ if $N' \subseteq N$ and $E' \subseteq E$.
- A tree $G_T = (N', T)$ of G is a connected and acyclic subgraph of G.
- $G_T = (N', T)$ is a spanning tree of G if it contains all nodes in G (ie, $N = N'$).
- The leaves of a tree are the nodes of degree 1.

Example

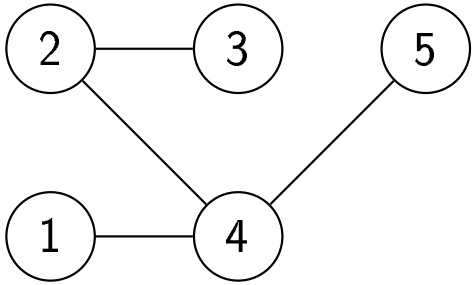




subgraph of G



tree of G



spanning tree of G

Properties of trees

Property 1

Every tree with n nodes has $n-1$ edges.

Proof.

- Base case: Holds for $n = 1$ (tree with 1 node and 0 edges).
- Inductive step: *Show that, if it holds for trees with n nodes, then it also holds for trees with $n+1$ nodes.*

  Let T_1 be any tree with $n+1$ nodes and recall that any tree with $n \geq 2$ nodes has at least 2 leaves.

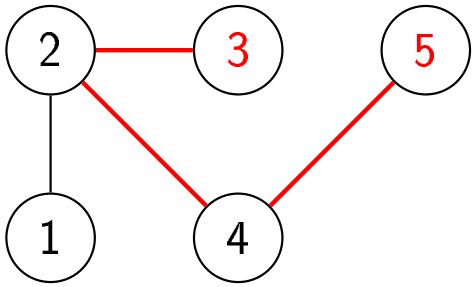
  By deleting one leaf and its incident edge, we obtain a tree T_2 with n nodes.

  By induction assumption, T_2 has $n-1$ edges. Thus, T_1 has $n-1+1 = n$ edges.

$\square$

Property 2

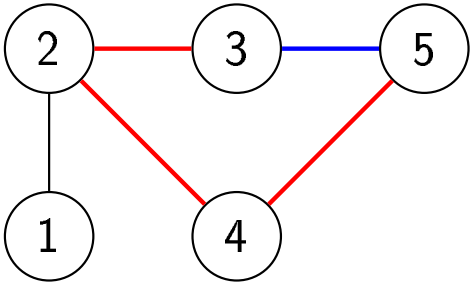
Any pair of nodes in a tree is connected via a unique path.



. . . otherwise there would be a cycle

Property 3

By adding a new edge to a tree, we create a unique cycle.

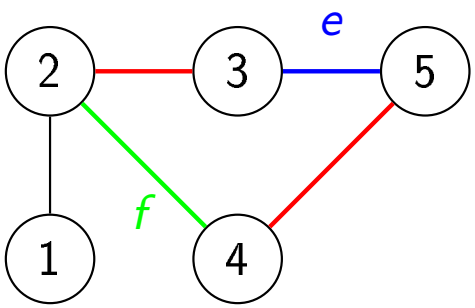


. . . consisting of the new edge (blue) and the corresponding unique path

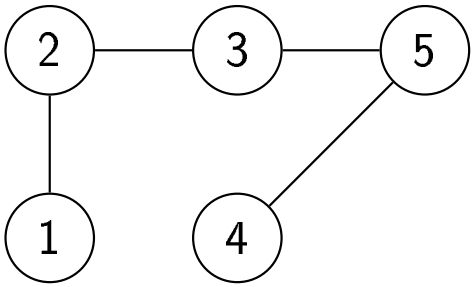
Exchange property

Property 4

Let $G_T = (N, T)$ be a spanning tree of $G = (N, E)$. Consider an edge $e \notin T$ and the unique cycle C of $T \cup \{e\}$.
For each edge $f \in C \backslash \{e\}$, the subgraph $T \cup \{e\} \backslash \{f\}$ is also a spanning tree of G.



G_T



$T \cup \{e\} \backslash \{f\}$